Lesson 9.3 - How to Calculate the Sample Size

> **Video:** How To Calculate The Sample Size Given The Confidence Level & Margin of Error by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Formulas

For a mean. Start from the margin of error:

$$E = z\,\frac{\sigma}{\sqrt{n}} \xrightarrow{\text{square both sides}} E^2 = \frac{z^2\sigma^2}{n} \;\Rightarrow\; \boxed{n = \frac{z^2\sigma^2}{E^2}}$$

For a proportion:

$$\boxed{n = \frac{z^2\,\hat{p}\,(1-\hat{p})}{E^2}}$$

If $\hat{p}$ is unknown, use $\hat{p} = 0.5$. That makes $\hat{p}(1-\hat{p})$ as large as possible, so the resulting n is big enough whatever the true proportion is.

Always **round** n **up** to the next whole number.

Example 1 - working backwards to the class size

The mean test score was 84 with standard deviation $\sigma = 15$. The margin of error at 95% confidence was $E = 5.367$. How many students were sampled?

$$n = \frac{(1.96)^2(15)^2}{(5.367)^2} \approx 30.0 \;\Rightarrow\; n = 30 \text{ students}$$

(The mean, 84, isn't needed.)

Example 2 - planning a survey on a proportion

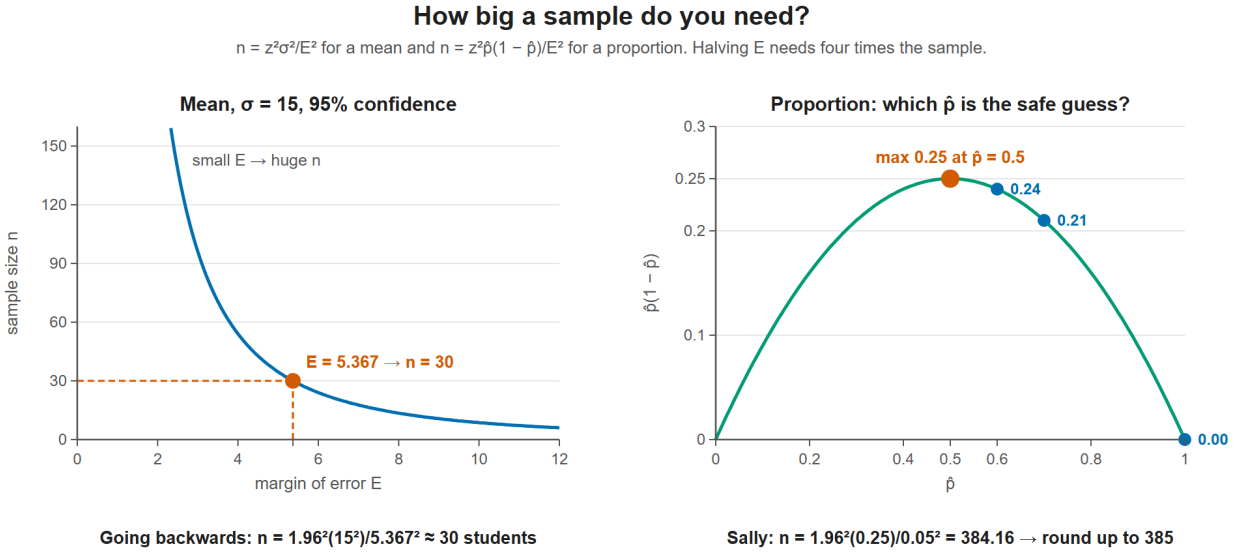
Sally wants to estimate the proportion of high-school students planning to go to college, with **95%** confidence and a **5%** margin of error ($E = 0.05$). She has no prior estimate of $\hat{p}$.

Which $\hat{p}$ makes $\hat{p}(1-\hat{p})$ largest?

$\hat{p}$	$1-\hat{p}$	**Product**
1.0	0.0	0.00
0.7	0.3	0.21
0.6	0.4	0.24
0.5	**0.5**	**0.25** (largest)

$$n = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2} = \frac{0.9604}{0.0025} = 384.16 \;\Rightarrow\; n = 385$$

Round **up**: 384 students would not quite reach the 5% margin.



Left: the required n grows fast as the margin of error shrinks, and Example 1's E = 5.367 corresponds to n = 30. Right: pÌ□(1 â□□ pÌ□) is largest at pÌ□ = 0.5, so using 0.5 when nothing is known gives the safest (largest) sample size.

Key takeaways

- Mean: $n = z^2\sigma^2/E^2$. Proportion: $n = z^2\hat{p}(1-\hat{p})/E^2$.
- With no estimate of $\hat{p}$, use 0.5. This gives the largest, safest n.
- Always round **up**.
- A smaller margin of error or a higher confidence level requires a larger sample.